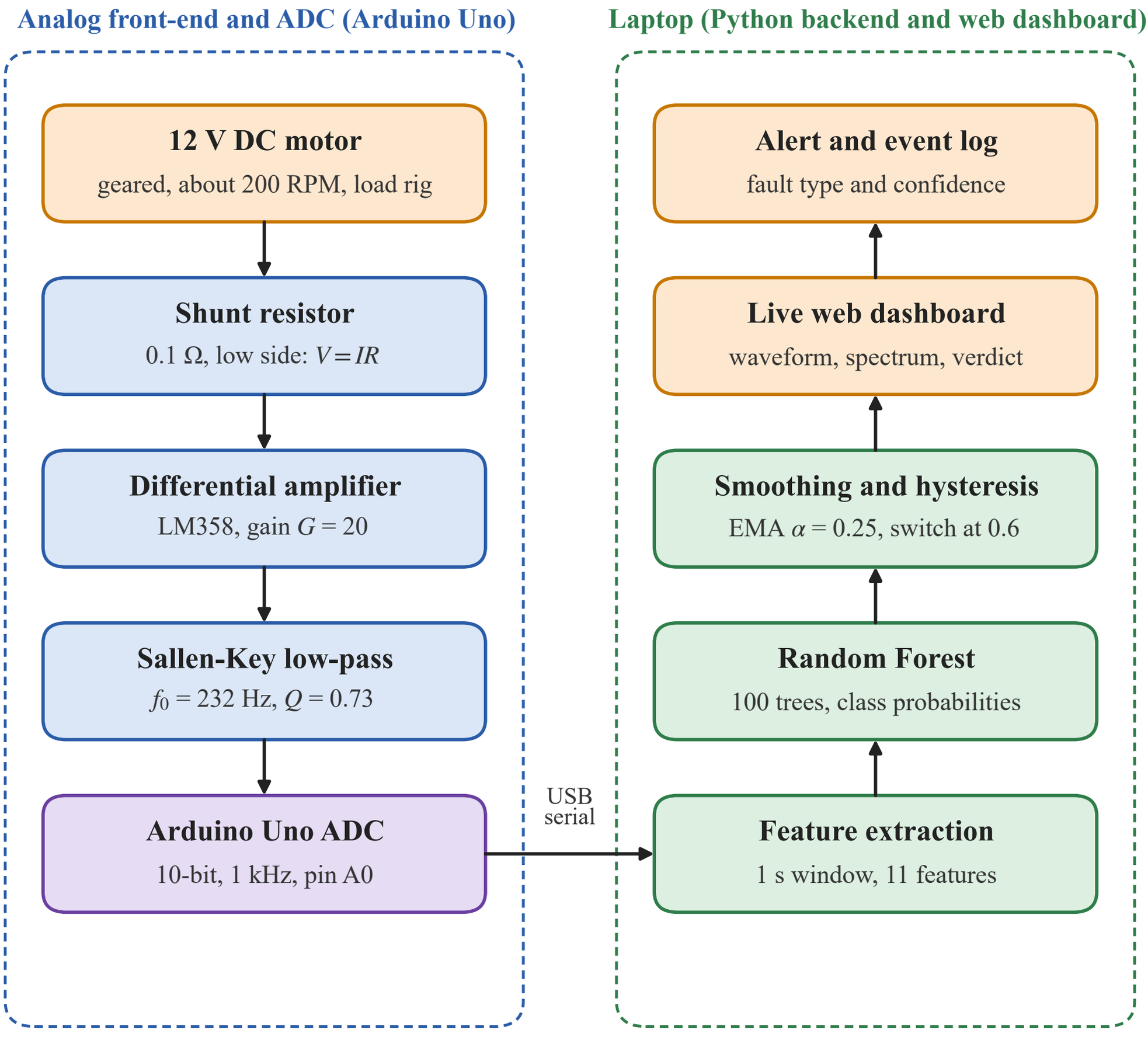

Analog front-end and ADC (Arduino Uno)
Laptop (Python backend and web dashboard)
12 V DC motor
geared, about 200 RPM, load rig
Shunt resistor
0.1 Ω, low side: V = IR
Differential amplifier
LM358, gain G = 20
Sallen-Key low-pass
f0 = 232 Hz, Q = 0.73
Arduino Uno ADC
10-bit, 1 kHz, pin A0
USB serial
Feature extraction
1 s window, 11 features
Random Forest
100 trees, class probabilities
Smoothing and hysteresis
EMA α = 0.25, switch at 0.6
Live web dashboard
waveform, spectrum, verdict
Alert and event log
fault type and confidence